Liang (Twist) Shan

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Research Interest

- Statistical foundations, optimization, and mechanistic interpretability of **Modern Machine Learning**, with a focus on deep learning, reinforcement learning, and generative AI, including language models and diffusion models.

Education

- **Peking University** — **Beijing, China**
  B.S. in Statistics, School of Mathematical Sciences — *Jun. 2024–Jul. 2028 (Expected)*
  Second Major B.A. in Economics, National School of Development — *Sep. 2025–Jul. 2028 (Expected)*
  - **Statistics Courses**: Probability Theory, Mathematical Statistics, Applied Stochastic Processes, Statistical Thinking, Stochastic Analysis and Applications,
  - **Computer Science Courses**: Data Structure and Algorithm, Fundamentals of Artificial Intelligence.
  - **Finance and Economics Courses**: Microeconomics, Macroeconomics, Econometrics (Honor), Mathematical Economics, Introduction to Finance, Security Investment, Mathematical Methods in Finance, Modern Quantitative Trading Systems, Topics in Quantitative Finance.
- **University of Copenhagen** — **Copenhagen, Denmark**
  International Exchange Student, Department of SCIENCE — *Sep. 2025–Jan. 2026*
  - **Mathematics Courses**: Topics in Statistics (MCMC), Advanced Operation Research: Stochastic Programming.
  - **Computer Science Courses**: Probability Machine Learning
- **Peking University** — **Beijing, China**
  B.S. in Physics, School of Earth and Space Sciences — *Sep. 2023–Jun. 2024*

Research Experiences

- **Mechanistic Interpretability of Chain-of-Thought Reasoning** — **University of Wisconsin–Madison**
  Advisor: Prof. Yiqiao Zhong, UW–Madison — *Apr. 2026 – Present*
  - Designed an interpretability pipeline combining probing, steering, ablation, and activation patching to identify and causally validate internal counting representations and their role in chain-of-thought reasoning.
  - Built synthetic counting tasks with known latent states to study how GPT-2-style Transformers encode running counts and compress sequence history.
  - Designed realistic needle-in-a-haystack prompts and localized the layers and token positions where open-weight LLMs track counts during chain-of-thought reasoning.
- **Pareto Frontier Model Identification from Pairwise Human Preferences** — **University of Pennsylvania**
  Advisor: Prof. Zhimei Ren, UPenn — *May. 2026 – Present*
  - Established theoretical guarantees for Pareto-set identification under Bradley–Terry feedback, leveraging Borda-score estimation to derive minimax sample-complexity bounds.
  - Designed synthetic scaling experiments and benchmarks, developed a tailored maximum-likelihood estimator, and built data-preprocessing and robustness-analysis pipelines for real-world pairwise comparisons.

Selected Course Projects

- **LLM Applications: WeChat Content Agent** — **Peking University**
  Advisor: Prof. Shanghang Zhang, PKU — *Apr. 2025 – Jun. 2025*
  - Built a Dify-based LLM agent that automated content generation, responsive HTML/CSS typesetting, and draft publishing through Python and WeChat APIs.
  - Integrated RAG and web-retrieval tools to support source-grounded writing and dynamic content collection.
- **Empirical Asset Pricing and Portfolio Management** — **Peking University**
  Advisor: Prof. Hai Huang, PKU — *Mar. 2025 – Jun. 2025*
  - Analyzed Chinese A-share data and constructed dynamically rebalanced portfolios using mean–variance optimization and fundamental signals.
  - Evaluated mutual-fund timing and stock-selection performance using the Treynor–Mazuy and Henriksson–Merton models.
  GitHub

- **Quantitative Trading Backtesting Framework** **Peking University**
  Advisor: Dr. Changhao Jiang, PKU *Sep. 2024 – Jan. 2025*
  - Developed a modular Python backtesting framework that decoupled signal generation, order execution, and performance evaluation.
  - Implemented and benchmarked momentum, mean-reversion, cross-sectional, and time-series strategies using vectorized data processing.

  GitHub

Selected Honors and Awards

- **Yu Hao Dream Scholarship** **Peking University**
  Scholarship supporting international exchange program *May. 2026*

Skills & Others

- **Computer Skills**: Python (PyTorch), R, C++ (Elementary), LaTeX
- **Languages**: Mandarin Chinese (Native), English (Fluent)
- **TOEFL iBT**: 105/120 (Reading 29, Listening 28, Speaking 23, Writing 25)
- **GRE General Test**: 325/340 (Verbal 155, Quantitative 170)